

ALGEBRA GENERAL EXAM - JANUARY 2026

Your UVa ID Number:

- There is a total of 100 points in 8 problems.
- Please provide *complete proofs* and justify *every* statement that you make.
- Make sure that the solution to every problem is a *continuous text* written legibly and in the correct order.
- If you refer to a *standard theorem*, please, *state* this theorem clearly and fully, and *justify* that this theorem applies in the situation at hand.
- Vague statements and hand-waving arguments will not be appreciated.
- You may assume the validity of one part of the problem to do another part even if you unable to prove the first part (in which case you, of course, will not get credit for it).

DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

Sign below the pledge:

“On my honor, I pledge that I have neither given nor received help on this assignment.”

Signature (Use your ID number instead): _____

- (1) (10 points) Let $p < q$ be primes such that $p \nmid q - 1$. Show that there is no group G such that $[G : Z(G)] = pq$, where $Z(G)$ is the center of G . (*Hint.* Consider the quotient $G/Z(G)$.)
- (2) Let $M = \mathbb{Z}^n$ considered as a $\mathbb{Z}$ -module. We say that a system of elements $x_1, \dots, x_r \in M$ can be extended to a basis of M if there exist $y_1, \dots, y_s \in M$ such that $x_1, \dots, x_r, y_1, \dots, y_s$ is a basis of M (then necessarily $r + s = n$).
 - (a) (7 points) Let $x_1, \dots, x_r \in M$ be linearly independent, and let $N \subset M$ be the $\mathbb{Z}$ -submodule generated by $x_1, \dots, x_r$. Show that if the quotient M/N is torsion-free then the system $x_1, \dots, x_r$ can be extended to a basis of M .
 - (b) (8 points) Let again $x_1, \dots, x_r \in M$, and let $N \subset M$ be the $\mathbb{Z}$ -submodule generated by $x_1, \dots, x_r$. Show that if the quotient M/N has nontrivial torsion (i.e. there exists $z \in M/N$, $z \neq 0$ such that $dz = 0$ for some integer $d > 0$) then the system $x_1, \dots, x_r$ cannot be extended to a basis of M .

- (3) Let $R = \mathbb{Q}[x]$ be the polynomial ring over $\mathbb{Q}$.

- (a) (5 points) Compute the tensor product of R -modules $M_1 \otimes_R M_2$, where

$$M_1 = R/(x^3 - 7x^2 + 14x - 8) \quad \text{and} \quad M_2 = R/(x^3 - 9x^2 + 23x - 15).$$

Express your answer as (a direct sum of) cyclic R -modules (i.e., (a direct sum of) modules of the form R/I for various ideals I).

- (b) (5 points) Let M be the module $M = \mathbb{Q}(x)$, the field of rational functions in x . Prove that the tensor product $M \otimes_R M$ is isomorphic to M .
- (c) (5 points) Consider the short exact sequence of R -modules:

$$0 \rightarrow R \xrightarrow{x} R \rightarrow R/(x) \rightarrow 0$$

where $x \cdot$ is multiplication by x (i.e., the map $r \mapsto xr$). Tensor this sequence with the module $N = R/(x^2)$ over R . Write down the resulting sequence of R -modules and maps, and determine if the resulting sequence is still exact.

- (4) Let R be the (commutative) ring of all continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ with the standard operations of addition and multiplication of functions.
 - (a) (2 points) Show that R is not an integral domain.
 - (b) (2 points) Find all units in R .
 - (c) (3 points) Show that if $f \in R$ satisfies $f^2 = f$ (i.e. f is an idempotent) then either $f \equiv 0$ or $f \equiv 1$, and deduce that R does not have a direct sum decomposition $R = R_1 \oplus R_2$ for nonzero rings R_1 and R_2 .
 - (d) (3 points) Show that the intersection of all prime ideals of R is (0) .
- (5) Let $f(x) = x^3 + 3x - 6$.
 - (a) (3 points) Show that $f(x)$ is irreducible over $\mathbb{Q}$.
 - (b) (3 points) Use (a) to show that $f(x)$ is also irreducible over $\mathbb{Q}(\sqrt{2026})$.
 - (c) (4 points) Let x_1, x_2, x_3 be the roots of f in $\mathbb{C}$. Find $x_1^2 + x_2^2 + x_3^2$.
- (6) Let K be a field, and let $A \in M_n(K)$ be a matrix. We say that A is diagonalizable over an extension L/K if there exists $Q \in GL_n(L)$ such that the matrix $Q^{-1}AQ$ is diagonal.
 - (a) (7 points) Let L/K be a finite Galois extension of degree $d = [L : K] > 1$. Show that for any $n \geq d$ there exists a matrix $A \in M_n(K)$ which is diagonalizable over L but is not diagonalizable over K .
 - (b) (8 points) Let E be a field extension of K , and let $K \subset L_1, L_2 \subset E$ be two intermediate subextensions. Show that if $A \in M_n(K)$ is diagonalizable over L_1 and L_2 then it is diagonalizable over $L_1 \cap L_2$.
- (7) We recall that the symmetric group $G = S_3$ has three irreducible complex representations: ρ_1 – the trivial representation, ρ_2 – the sign representation, and ρ_3 – the 2-dimensional representation given by

$$\rho_3((12)) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad \rho_3((123)) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}.$$

Let us now consider the matrix algebra $V = M_2(\mathbb{C})$ as a 4-dimensional vector space, and define an action of $g \in G$ on V by

$$\rho(g)(A) = \rho_3(g)A\rho_3(g)^{-1} \quad \text{for } A \in M_2(\mathbb{C}).$$

- (a) (5 points) Show that $\rho(g_1g_2) = \rho(g_1) \cdot \rho(g_2)$ (composition) for $g_1, g_2 \in G$; in other words, we have a representation $\rho: G \rightarrow GL_4(\mathbb{C})$. (You can assume that ρ_3 is a representation.)
 - (b) (5 points) Find the values of the character of ρ on all three conjugacy classes of G with representatives (1) , (12) , and (123) .
 - (c) (5 points) Find the decomposition of ρ as a sum of irreducibles.
- (8) (10 points) Let K be a field, $f(x) \in K[x]$ be an irreducible separable polynomial, and let L be the splitting field of f over K . Assume that the Galois group $\text{Gal}(L/K)$ is abelian. Show that every root α of f is a primitive element for L over K , i.e. $L = K(\alpha)$.